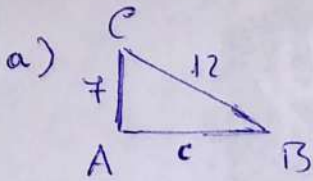


Ficha de repaso

Soluciones

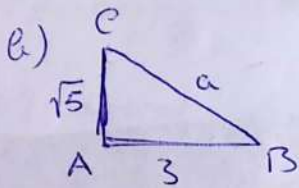
1.-



$$\sin B = \frac{7}{12} \Rightarrow B = \arcsin \frac{7}{12} = 35'69''$$

$$\cos C = \frac{7}{12} \Rightarrow C = \arccos \frac{7}{12} = 54'31''$$

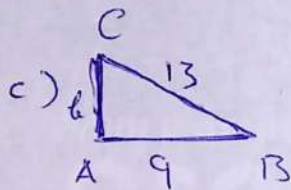
$$7^2 + c^2 = 12^2 \Rightarrow c = \sqrt{12^2 - 7^2} = \sqrt{95} \text{ cm.}$$



$$\tan B = \frac{\sqrt{5}}{3} \Rightarrow B = \arctan \frac{\sqrt{5}}{3} = 36'7''$$

$$\cot C = \frac{\sqrt{5}}{3} \Rightarrow C = \operatorname{arctg} \frac{3}{\sqrt{5}} = 53'30''$$

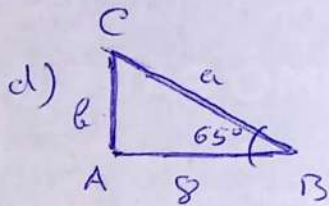
$$a^2 = (\sqrt{5})^2 + 3^2 \Rightarrow a = \sqrt{5+9} = \sqrt{14} \text{ cm.}$$



$$\cos B = \frac{9}{13} \Rightarrow B = \arccos \frac{9}{13} = 46'19''$$

$$C = \arcsin \frac{9}{13} = 43'81''$$

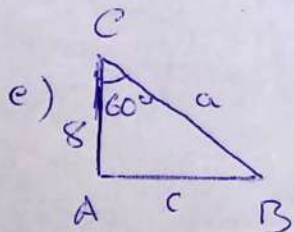
$$b = \sqrt{13^2 - 9^2} = \sqrt{88} \text{ cm}$$



$$\tan 65^\circ = \frac{b}{8} \Rightarrow b = 8 \cdot \tan 65^\circ = 17'16 \text{ cm.}$$

$$8^2 + 17'16^2 = a^2 \Rightarrow a = \sqrt{358'47} = 18'93 \text{ cm.}$$

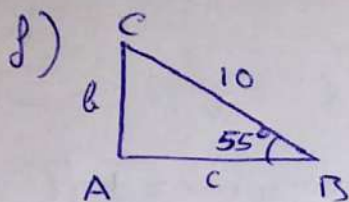
$$C = 180^\circ - (90^\circ + 65^\circ) = 25^\circ$$



$$\tan 60^\circ = \frac{c}{8} \Rightarrow c = 8 \cdot \tan 60^\circ = 13'86 \text{ cm}$$

$$\cos 60^\circ = \frac{8}{a} \Rightarrow a = \frac{8}{\cos 60^\circ} = 16 \text{ cm.}$$

$$B = 180^\circ - (90^\circ + 60^\circ) = 30^\circ$$



$$\sin 55^\circ = \frac{b}{10} \Rightarrow b = 10 \cdot \sin 55^\circ = 8'19 \text{ cm}$$

$$\cos 55^\circ = \frac{c}{10} \Rightarrow c = 10 \cdot \cos 55^\circ = 5'74 \text{ cm}$$

$$C = 180^\circ - (90^\circ + 55^\circ) = 35^\circ$$

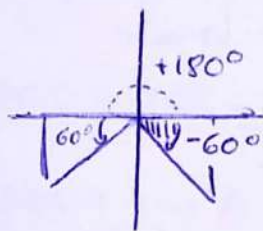
2.- a) $\arcsin \frac{\sqrt{2}}{2} = 45^\circ \rightarrow$ Como α está en el segundo cuadrante $\Rightarrow \alpha = 180^\circ - 45^\circ = 135^\circ = \frac{3\pi}{4} \text{ rad.}$

b) $\text{arctg}(-1) = -45^\circ \rightarrow 4^\circ \text{ cuadrante} \rightarrow \alpha = 360^\circ - 45^\circ = 315^\circ \left(\frac{7\pi}{4} \text{ rad} \right)$

c) $\sec \alpha = -\frac{2\sqrt{3}}{3} \Rightarrow \cos \alpha = \frac{1}{\sec \alpha} = -\frac{3}{2\sqrt{3}}$
 $\Rightarrow \arccos \left(+\frac{3}{2\sqrt{3}} \right) = 30^\circ \Rightarrow 3^\circ \text{ cuadrante: } \alpha = 180^\circ + 30^\circ = \underline{210^\circ}$

d) $\arccos 0 = 90^\circ \Rightarrow 3^\circ \text{ cuadrante} \Rightarrow 180^\circ + 90^\circ = \underline{270^\circ}$

e) $\arcsin \left(-\frac{\sqrt{3}}{2} \right) = -60^\circ$ m) al meter el ángulo negativo me da la solución del 4.º cuadrante. Como queremos el 3.º será $\rightarrow \alpha = 180^\circ + 60^\circ = 240^\circ$.



f) $\text{cosec} \alpha = -2 \Rightarrow \sin \alpha = \frac{1}{\text{cosec} \alpha} = -\frac{1}{2} \Rightarrow \arcsin \left(-\frac{1}{2} \right) = -30^\circ$

Igual que antes, para sacar 3.º cuadrante $\rightarrow \alpha = 180^\circ + 30^\circ = \underline{210^\circ}$.

$$(3.-) a) \operatorname{tg} \alpha = -\sqrt{5} \text{ y } \frac{\pi}{2} < \alpha \leq \pi$$

$$\Rightarrow 1 + \operatorname{tg}^2 \alpha = \sec^2 \alpha \Rightarrow 1 + (-\sqrt{5})^2 = \sec^2 \alpha \Rightarrow \sec \alpha = \pm \sqrt{6}$$

$$\cos \alpha = \frac{1}{\sec \alpha} = \frac{1}{-\sqrt{6}} = -\frac{\sqrt{6}}{6} \text{ (racionalizando). } \downarrow \text{ 2.º cuadrante}$$

$$\operatorname{tg} \alpha = \frac{\operatorname{sen} \alpha}{\cos \alpha} \Rightarrow \operatorname{sen} \alpha = \operatorname{tg} \alpha \cdot \cos \alpha = (-\sqrt{5}) \cdot \left(-\frac{\sqrt{6}}{6}\right) = \frac{\sqrt{30}}{6}$$

$$\operatorname{cosec} \alpha = \frac{1}{\operatorname{sen} \alpha} = \frac{1}{\frac{\sqrt{30}}{6}} = \frac{6}{\sqrt{30}} = \frac{6\sqrt{30}}{30} = \frac{\sqrt{30}}{5}$$

$$\operatorname{cotg} \alpha = \frac{1}{\operatorname{tg} \alpha} = \frac{1}{-\sqrt{5}} = -\frac{\sqrt{5}}{5}$$

$$(4.) \operatorname{cosec} \alpha = -\frac{5}{4} \text{ y } \frac{3\pi}{2} < \alpha \leq 2\pi$$

$$\Rightarrow \operatorname{sen} \alpha = \frac{1}{\operatorname{cosec} \alpha} = \frac{1}{-\frac{5}{4}} = -\frac{4}{5}$$

$$\operatorname{sen}^2 \alpha + \cos^2 \alpha = 1 \Rightarrow \left(-\frac{4}{5}\right)^2 + \cos^2 \alpha = 1 \Rightarrow \cos \alpha = \pm \sqrt{1 - \frac{16}{25}}$$

$$\Rightarrow \cos \alpha = \pm \sqrt{\frac{9}{25}} = \pm \frac{3}{5}$$

↓
4.º cuadrante

$$\sec \alpha = \frac{1}{\cos \alpha} = \frac{1}{\frac{3}{5}} = \frac{5}{3}$$

$$\operatorname{tg} \alpha = \frac{\operatorname{sen} \alpha}{\cos \alpha} = \frac{-\frac{4}{5}}{\frac{3}{5}} = -\frac{4}{3} \Rightarrow \operatorname{cotg} \alpha = \frac{1}{\operatorname{tg} \alpha} = -\frac{3}{4}$$

c)

$$\cotg \alpha = \sqrt{7} \text{ y } \pi < \alpha \leq \frac{3\pi}{2}$$

$$\tg \alpha = \frac{1}{\cotg \alpha} = \frac{1}{\sqrt{7}} = \frac{\sqrt{7}}{7}$$

$$1 + \tg^2 \alpha = \sec^2 \alpha \Rightarrow 1 + \left(\frac{\sqrt{7}}{7}\right)^2 = \sec^2 \alpha \Rightarrow 1 + \frac{1}{7} = \sec^2 \alpha$$

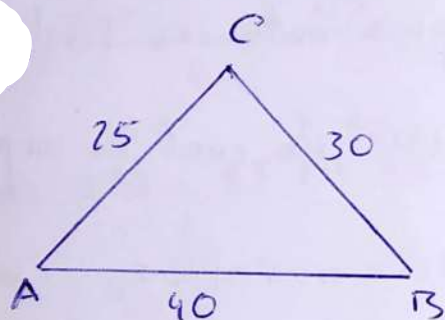
$$\Rightarrow \frac{8}{7} = \sec^2 \alpha \Rightarrow \sec \alpha = \pm \sqrt{\frac{8}{7}} \Rightarrow \cos \alpha = \frac{1}{\sec \alpha} = -\sqrt{\frac{7}{8}}$$

3.º cuadrante.

$$\tg \alpha = \frac{\sen \alpha}{\cos \alpha} \Rightarrow \sen \alpha = \tg \alpha \cdot \cos \alpha = \frac{\sqrt{7}}{7} \cdot \left(-\sqrt{\frac{7}{8}}\right) = -\frac{7}{7\sqrt{8}} = -\frac{1}{\sqrt{8}} = -\frac{\sqrt{8}}{8}$$

$$\operatorname{cosec} \alpha = \frac{1}{\sen \alpha} = -\frac{8}{\sqrt{8}} = -\sqrt{8}$$

4.-



Como conocemos los tres lados vamos a usar el T. del coseno $\Rightarrow$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\Rightarrow 30^2 = 25^2 + 40^2 - 2 \cdot 25 \cdot 40 \cdot \cos A$$

$$\Rightarrow 900 = 625 + 1600 - 2000 \cos A \Rightarrow \cos A =$$

$$\Rightarrow 900 = 2225 - 2000 \cdot \cos A \Rightarrow \cos A = \frac{2225 - 900}{2000} = 0'6625$$

$$\Rightarrow \text{arccos } 0'6625 = 48'51''$$

Ahora bien, hay dos ángulos cuyo coseno vale 0'6625, son $48'51''$ y $360^\circ - 48'51'' = 311'49''$. Como en un triángulo no puede haber ángulos mayores de 180° , nos quedamos con $48'51''$.

Por tanto $A = 48'51''$

Ahora que sabemos 1 ángulo, podemos hallar B con el teorema del seno.

$$\Rightarrow \frac{a}{\text{sen } A} = \frac{b}{\text{sen } B} \Rightarrow \frac{30}{\text{sen } 48'51''} = \frac{25}{\text{sen } B}$$

$$\Rightarrow \text{sen } B = \frac{25 \cdot \text{sen } 48'51''}{30} \approx 0'6242$$

$$\arcsen 0'6242 = 38'63''$$

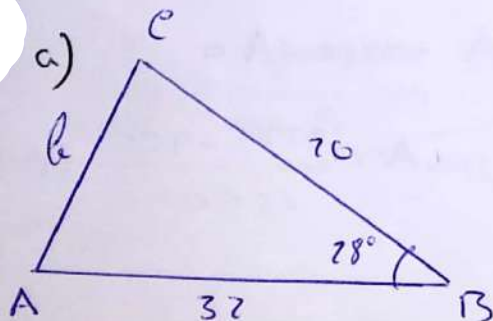
Hay dos ángulos cuyo seno vale 0'6242, $\text{sen } 38'63''$ y $180^\circ - 38'63'' = 141'37''$.

Ahora bien, $141'37''$ no es válido pues entonces tendríamos

$A + B = 48'51'' + 141'37'' = 189'88'' > 180^\circ$ lo cual es imposible. Por tanto $B = 38'63''$

$$\text{Para hallar } C \Rightarrow C = 180^\circ - (48'51'' + 38'63'') = \boxed{92'86''}$$

5.-



Teorema del coseno:

$$b^2 = 32^2 + 20^2 - 2 \cdot 32 \cdot 20 \cdot \cos 28^\circ$$

$$b^2 = 293'83 \Rightarrow \boxed{b = 17'14 \text{ cm}}$$

$$\text{Teorema del seno} \rightarrow \frac{20}{\text{sen } A} = \frac{17'14}{\text{sen } 28^\circ} = \frac{32}{\text{sen } C}$$

$$\Rightarrow \text{sen } A = \frac{20 \cdot \text{sen } 28^\circ}{17'14} = 0'55 \Rightarrow \arcsen 0'55 = 32'72''$$

$$A = 32'72''$$

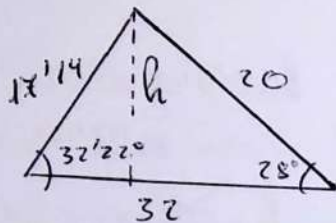
$$B = 180^\circ - 32'72'' - 92'86'' = 146'49''$$

Únicamente tiene sentido la solución $\boxed{A = 32'22^\circ}$

$$\Rightarrow C = 180^\circ - (28^\circ + 32'22^\circ) = \boxed{119'78^\circ}$$

Para calcular el área necesitamos la altura h .

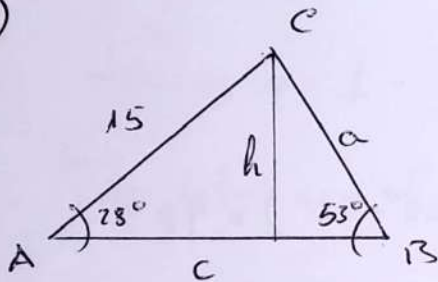
$\Rightarrow$



$$\text{sen } 28^\circ = \frac{h}{20} \Rightarrow h = 20 \cdot \text{sen } 28^\circ = 9'39 \text{ cm}$$

$$A = \frac{b \cdot h}{2} = \frac{32 \cdot 9'39}{2} = 150'23 \text{ cm}^2$$

b)



$$\frac{a}{\text{sen } 28^\circ} = \frac{15}{\text{sen } 53^\circ} \Rightarrow a = \frac{15 \cdot \text{sen } 28^\circ}{\text{sen } 53^\circ} = \boxed{8'82 \text{ cm}}$$

$$15^2 = a^2 + c^2 - 2 \cdot a \cdot c \cdot \cos 53^\circ$$

$$15^2 = 8'82^2 + c^2 - 2 \cdot 8'82 \cdot c \cdot \cos 53^\circ$$

$$\Rightarrow 225 = 77'79 + c^2 - 10'62 \cdot c$$

$$\Rightarrow c^2 - 10'62c - 147'21 = 0$$

$$c = \frac{10'62 \pm \sqrt{10'62^2 - 4 \cdot 1 \cdot (-147'21)}}{2 \cdot 1} = \frac{10'62 \pm 26'49}{2}$$

$$\boxed{18'55 \text{ cm}}$$

$$-7'94 \text{ cm}$$

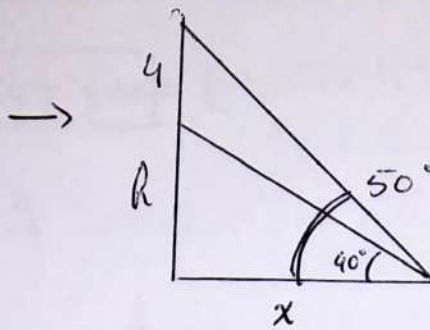
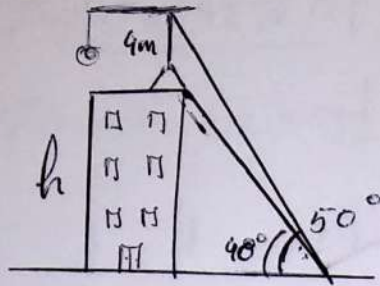
$$\Rightarrow \frac{c}{\text{sen } C} = \frac{b}{\text{sen } B} \Rightarrow \frac{18'55}{\text{sen } C} = \frac{15}{\text{sen } 53^\circ}$$

$$C = 180^\circ - (28^\circ + 53^\circ) = \boxed{99^\circ}$$

$$\Rightarrow h = 15 \cdot \text{sen } 28^\circ = 7'04 \text{ cm}$$

$$\Rightarrow A = \frac{b \cdot h}{2} = \frac{18'55 \cdot 7'04}{2} = \boxed{65'3 \text{ cm}^2}$$

6.-



$$\Rightarrow \left. \begin{aligned} \operatorname{tg} 50^\circ &= \frac{4+h}{x} \\ \operatorname{tg} 40^\circ &= \frac{h}{x} \end{aligned} \right\}$$

$$\Rightarrow \left. \begin{aligned} 1.91x &= 4+h \\ 0.84x &= h \end{aligned} \right\} -$$

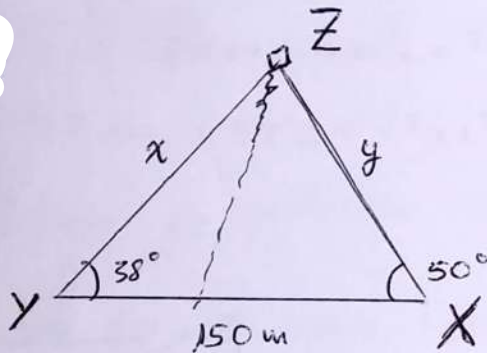
$$h = 0.84x = 0.84 \cdot 11.37 = 9.55 \text{ m}$$

$$\frac{h}{0.35} = 4 \Rightarrow x = \frac{4}{0.35} = 11.37 \text{ m}$$

$$\Rightarrow h = 0.84x = 0.84 \cdot 11.37 = 9.55 \text{ m}$$

El edificio mide 9.55 m.

7.-



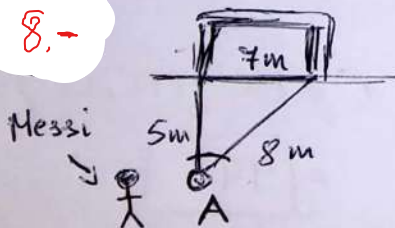
$$Z = 180^\circ - (38^\circ + 50^\circ) = 92^\circ$$

$$\frac{x}{\operatorname{sen} 50^\circ} = \frac{y}{\operatorname{sen} 38^\circ} = \frac{150}{\operatorname{sen} 92^\circ}$$

$$\Rightarrow x = \frac{150 \operatorname{sen} 50^\circ}{\operatorname{sen} 92^\circ} = 114.98 \text{ m}$$

$$y = \frac{150 \operatorname{sen} 38^\circ}{\operatorname{sen} 92^\circ} = 92.40 \text{ m}$$

8.-



Hay que hallar A en la situación anterior
 $\Rightarrow$ Aplicamos el T. del coseno.

$$7^2 = 5^2 + 8^2 - 2 \cdot 5 \cdot 8 \cdot \cos A \Rightarrow 49 = 89 - 80 \cos A$$

$$\Rightarrow \cos A = \frac{89 - 49}{80} = \frac{1}{2} \Rightarrow \arccos \frac{1}{2} = 60^\circ \left\{ \begin{aligned} A &= 60^\circ \text{ sí} \\ A &= 360^\circ - 60^\circ = 300^\circ \text{ No} \end{aligned} \right.$$

Messi tiene un ángulo de 60° para golpear el balón.

9.- Información que tengo $\rightarrow \sin \alpha = \frac{4}{5}$, $0 \leq \alpha \leq 90^\circ$

Fórmulas que voy a utilizar $\rightarrow$

$$\left\{ \begin{array}{l} \sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha \\ \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha \\ \sin(180^\circ + \alpha) = -\sin \alpha \\ \cos(180^\circ - \alpha) = -\cos \alpha \end{array} \right.$$

¿Qué necesito para aplicar las fórmulas anteriores?
 $\rightarrow$ Calcular $\cos \alpha$.

$$\Rightarrow \sin^2 \alpha + \cos^2 \alpha = 1 \Rightarrow \left(\frac{4}{5}\right)^2 + \cos^2 \alpha = 1$$

$$\Rightarrow \cos^2 \alpha = 1 - \frac{16}{25} \Rightarrow \cos \alpha = \pm \sqrt{\frac{9}{25}} = \pm \frac{3}{5}$$

Por estar en el 1.º cuadrante $\Rightarrow \cos \alpha = +\frac{3}{5}$

Ahora ya podemos aplicar las fórmulas:

$$a) \sin 2\alpha = 2 \cdot \sin \alpha \cdot \cos \alpha = 2 \cdot \frac{4}{5} \cdot \frac{3}{5} = \frac{24}{25}$$

$$b) \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$$

$$c) \sin(180^\circ + \alpha) = -\sin \alpha = -\frac{4}{5}$$

$$d) \cos(180^\circ - \alpha) = -\cos \alpha = -\frac{3}{5}$$

De estos dos
signos deduci-
mos que 2α
está en el 2º
cuadrante

(como curiosidad)

10.- $\operatorname{tg} \alpha = 2$, $\cos \alpha > 0$. m, $\operatorname{tg} > 0$ y $\cos > 0 \Rightarrow$ estoy en el 1.º cuadrante.

a) $\hookrightarrow \boxed{\cos 2\alpha = \cos^2 \alpha - \operatorname{sen}^2 \alpha} \Rightarrow$ Necesito $\cos \alpha$ y $\operatorname{sen} \alpha$. Pues $\cos \alpha > 0$.

$$\Rightarrow \begin{cases} 1 + \operatorname{tg}^2 \alpha = \frac{1}{\cos^2 \alpha} \Rightarrow 1 + 2^2 = \frac{1}{\cos^2 \alpha} \Rightarrow \cos^2 \alpha = \frac{1}{5} \Rightarrow \cos \alpha = \pm \sqrt{\frac{1}{5}} \\ \operatorname{sen}^2 \alpha + \cos^2 \alpha = 1 \Rightarrow \operatorname{sen}^2 \alpha = 1 - \left(\sqrt{\frac{1}{5}}\right)^2 = 1 - \frac{1}{5} = \frac{4}{5} \Rightarrow \operatorname{sen} \alpha = \pm \sqrt{\frac{4}{5}} = \end{cases}$$

1.º cuadrante $\Rightarrow \cos \alpha = \frac{+2}{\sqrt{5}}$
~~No sé con~~
~~cual me quedo.~~

$\Rightarrow \cos 2\alpha = \cos^2 \alpha - \operatorname{sen}^2 \alpha = \left(\sqrt{\frac{1}{5}}\right)^2 - \left(\frac{2}{\sqrt{5}}\right)^2 = \frac{1}{5} - \frac{4}{5} = \boxed{\frac{-3}{5}}$

b) $\operatorname{sen}(\pi - \alpha) = \operatorname{sen} \alpha = \frac{2}{\sqrt{5}}$

c) $\operatorname{tg} 2\alpha = \frac{2 \operatorname{tg} \alpha}{1 - \operatorname{tg}^2 \alpha} = \frac{2 \cdot 2}{1 - 2^2} = \frac{4}{-3} = -\frac{4}{3}$

d) $\operatorname{tg}\left(\frac{\pi}{4} + \alpha\right) = \frac{\operatorname{tg} \frac{\pi}{4} + \operatorname{tg} \alpha}{1 - \operatorname{tg} \frac{\pi}{4} \cdot \operatorname{tg} \alpha} = \frac{1 + 2}{1 - 1 \cdot 2} = \frac{3}{-1} = -3$

11.-

$$a) \cos x = 0 \rightarrow \arccos 0 = 90^\circ \begin{cases} \nearrow x = 90^\circ + 360^\circ \cdot K \\ \searrow x = 270^\circ + 360^\circ \cdot K \end{cases}$$

$$b) \operatorname{tg} x = -1 \rightarrow \operatorname{arctg}(-1) = -45^\circ \begin{cases} \nearrow x = 315^\circ + 360^\circ \cdot K \\ \searrow x = 135^\circ + 360^\circ \cdot K \end{cases}$$

$360^\circ - 45^\circ = 315^\circ$

$$c) 2 \cdot \operatorname{cosec} x - 4 = 0$$

$$\operatorname{cosec} x = \frac{4}{2} = 2 \rightarrow \operatorname{sen} x = \frac{1}{2} \rightarrow \operatorname{arcsen} \frac{1}{2} = 30^\circ$$

$$\begin{cases} \nearrow x = 30^\circ + 360^\circ \cdot K \\ \searrow x = 150^\circ + 360^\circ \cdot K \end{cases}$$

$$d) \cot^2 x = 4 \rightarrow \cot x = \pm \sqrt{4} = \pm 2 \Rightarrow \operatorname{tg} x = \pm \frac{1}{2}$$

$$\bullet \text{ Si } \operatorname{tg} x = \frac{1}{2} \rightarrow \operatorname{arctg} \frac{1}{2} = 26'57'' \begin{cases} \nearrow x = 26'57'' + 360^\circ \cdot K \\ \searrow x = 206'57'' + 360^\circ \cdot K \end{cases}$$

$$\bullet \text{ Si } \operatorname{tg} x = -\frac{1}{2} \rightarrow \operatorname{arctg}\left(-\frac{1}{2}\right) = -26'57'' \begin{cases} \nearrow x = 333'43'' + 360^\circ \cdot K \\ \searrow x = 153'43'' + 360^\circ \cdot K \end{cases}$$

$$e) \operatorname{sen}^2 x - 1 = 0 \rightarrow \operatorname{sen} x = \pm 1$$

$$\bullet \text{ Si } \operatorname{sen} x = 1 \Rightarrow \operatorname{arcsen} 1 = 90^\circ \Rightarrow x = 90^\circ + 360^\circ \cdot K$$

$$\bullet \text{ Si } \operatorname{sen} x = -1 \Rightarrow \operatorname{arcsen}(-1) = 270^\circ \Rightarrow x = 270^\circ + 360^\circ \cdot K$$

$$f) \operatorname{sen} x + \cos x = 0 \rightarrow \frac{\operatorname{sen} x}{\cos x} + \frac{\cos x}{\cos x} = \frac{0}{\cos x}$$

$$\operatorname{tg} x + 1 = 0 \rightarrow \operatorname{tg} x = -1$$

$$\operatorname{arctg}(-1) = -45^\circ \Rightarrow \begin{cases} \nearrow x = 315^\circ + 360^\circ \cdot K \\ \searrow x = 135^\circ + 360^\circ \cdot K \end{cases}$$

$$g) \operatorname{tg} x = \sqrt{2} \cdot \cos x$$

$$\operatorname{sen}^2 x + \cos^2 x = 1$$

$$\frac{\operatorname{sen} x}{\cos x} = \sqrt{2} \cdot \cos x \Rightarrow \operatorname{sen} x = \sqrt{2} \cdot \cos^2 x \xRightarrow{\text{}} \operatorname{sen} x = \sqrt{2} (1 - \operatorname{sen}^2 x)$$

$$\Rightarrow \sqrt{2} \cdot \operatorname{sen}^2 x + \operatorname{sen} x - \sqrt{2} = 0 \quad \rightarrow \boxed{y = \operatorname{sen} x}$$

$$\sqrt{2} \cdot y^2 + y - \sqrt{2} = 0$$

$$y = \frac{-1 \pm \sqrt{1^2 - 4 \cdot \sqrt{2} \cdot (-\sqrt{2})}}{2 \cdot \sqrt{2}} = \frac{-1 \pm \sqrt{1+8}}{2\sqrt{2}} = \frac{-1 \pm 3}{2\sqrt{2}} \quad \left| \begin{array}{l} \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \\ \frac{-4}{2\sqrt{2}} = \frac{-2}{\sqrt{2}} = -\sqrt{2} \end{array} \right.$$

• Si $y = \frac{\sqrt{2}}{2} \Rightarrow \frac{\sqrt{2}}{2} = \operatorname{sen} x \Rightarrow \arcsen \frac{\sqrt{2}}{2} = 45^\circ \quad \left\{ \begin{array}{l} x = 45^\circ + 360^\circ K \\ x = 135^\circ + 360^\circ K \end{array} \right.$

• Si $y = -\sqrt{2} \Rightarrow -\sqrt{2} = \operatorname{sen} x \Rightarrow$ No hay solución.

$$h) \cos\left(\frac{\pi}{6} + x\right) = \operatorname{sen} x$$

↓

$$\cos \frac{\pi}{6} \cdot \cos x - \operatorname{sen} \frac{\pi}{6} \cdot \operatorname{sen} x = \operatorname{sen} x$$

$$\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \operatorname{sen} x = \operatorname{sen} x \rightarrow \frac{\sqrt{3}}{2} \cos x = \operatorname{sen} x + \frac{1}{2} \operatorname{sen} x$$

$$\rightarrow \frac{\sqrt{3}}{2} \cos x = \frac{3}{2} \operatorname{sen} x \rightarrow \sqrt{3} \cos x = 3 \operatorname{sen} x \rightarrow$$

$$\rightarrow \sqrt{3} \frac{\cos x}{\cos x} = 3 \frac{\operatorname{sen} x}{\cos x} \rightarrow \sqrt{3} = 3 \operatorname{tg} x \rightarrow \operatorname{tg} x = \frac{\sqrt{3}}{3}$$

$$\arctg \frac{\sqrt{3}}{3} = 30^\circ \quad \left\{ \begin{array}{l} x = 30^\circ + 360^\circ K \\ x = 210^\circ + 360^\circ K \end{array} \right.$$

$$i) \quad \operatorname{tg} 2x + 2\cos x = 0 \rightarrow \operatorname{tg} 2x = -2\cos x$$

$$\rightarrow \frac{2\operatorname{tg} x}{1 - \operatorname{tg}^2 x} = -2\cos x \rightarrow \cancel{2}\operatorname{tg} x = -\cancel{2}\cos x (1 - \operatorname{tg}^2 x)$$

$$\rightarrow \frac{\operatorname{sen} x}{\cos x} = \cos x \left(1 - \frac{\operatorname{sen}^2 x}{\cos^2 x}\right)$$

$$\rightarrow \operatorname{sen} x = \cos^2 x \left(1 - \frac{\operatorname{sen}^2 x}{\cos^2 x}\right)$$

$$\rightarrow \operatorname{sen} x = \cos^2 x - \cancel{\cos^2 x} \cdot \frac{\operatorname{sen}^2 x}{\cancel{\cos^2 x}}$$

$$\rightarrow \operatorname{sen} x = \cos^2 x - \operatorname{sen}^2 x \rightarrow \text{Usamos que } \cos^2 x + \operatorname{sen}^2 x = 1$$

$$\cos^2 x = 1 - \operatorname{sen}^2 x$$

$$\Rightarrow \operatorname{sen} x = (1 - \operatorname{sen}^2 x) - \operatorname{sen}^2 x$$

$$\rightarrow \operatorname{sen} x = 1 - 2\operatorname{sen}^2 x \rightarrow \boxed{2\operatorname{sen}^2 x + \operatorname{sen} x - 1 = 0}$$

$$C) \quad \textcircled{y = \operatorname{sen} x} \rightarrow 2y^2 + y - 1 = 0 \rightarrow y = \frac{-1 \pm \sqrt{9}}{4} = \frac{-1 \pm 3}{4} \quad \begin{matrix} 1/2 \\ -4/4 = -1 \end{matrix}$$

$$\bullet \text{ Si } y = 1/2 \Rightarrow \operatorname{sen} x = 1/2 \Rightarrow \begin{cases} x = 30^\circ + 360^\circ K \\ x = 150^\circ + 360^\circ K \end{cases}$$

$$\bullet \text{ Si } y = -1 \Rightarrow \operatorname{sen} x = -1 \Rightarrow x = 270^\circ + 360^\circ K$$

$$j) \quad 3\operatorname{sen}^2 x + \cos^2 x + \cos x = 0 \rightarrow 3(1 - \cos^2 x) + \cos^2 x + \cos x = 0$$

$$\cancel{3\operatorname{sen}^2 x + 3(1 - \operatorname{sen}^2 x) + \cos x} = 0 \quad 3 - 3\cos^2 x + \cos^2 x + \cos x = 0$$

$$\Rightarrow -2\cos^2 x + \cos x + 3 = 0 \rightarrow -2y^2 + y + 3 = 0$$

$$y = \frac{-1 \pm \sqrt{25}}{-4} = \frac{-1 \pm 5}{-4} \quad \begin{matrix} \frac{4}{-4} = -1 \rightarrow \cos x = -1 \rightarrow x = 180^\circ + 360^\circ K \\ \frac{-6}{-4} = \frac{3}{2} \rightarrow \cos x = \frac{3}{2} \rightarrow \text{No sol.} \end{matrix}$$

$$12.- a) \frac{\operatorname{sen} 2\alpha}{1 - \cos^2 \alpha} = \frac{2 \operatorname{sen} \alpha \cdot \cos \alpha}{1 - \cos^2 \alpha} = \frac{2 \operatorname{sen} \alpha \cos \alpha}{\operatorname{sen}^2 \alpha} = \frac{2 \cos \alpha}{\operatorname{sen} \alpha} =$$

$$= 2 \cotg \alpha$$

$$b) \frac{2 \cos(45^\circ + \alpha) \cos(45^\circ - \alpha)}{\cos 2\alpha} =$$

$$= \frac{2 \cdot (\cos 45^\circ \cos \alpha - \operatorname{sen} 45^\circ \operatorname{sen} \alpha) \cdot (\cos 45^\circ \cos \alpha + \operatorname{sen} 45^\circ \operatorname{sen} \alpha)}{\cos 2\alpha} =$$

$$= \frac{2 \left(\frac{\sqrt{2}}{2} \cos \alpha - \frac{\sqrt{2}}{2} \operatorname{sen} \alpha \right) \cdot \left(\frac{\sqrt{2}}{2} \cos \alpha + \frac{\sqrt{2}}{2} \operatorname{sen} \alpha \right)}{\cos^2 \alpha - \operatorname{sen}^2 \alpha} \quad \begin{array}{l} \text{Suma por} \\ \uparrow \text{diferencia} \end{array}$$

$$= \frac{2 \left(\left(\frac{\sqrt{2}}{2} \cos \alpha \right)^2 - \left(\frac{\sqrt{2}}{2} \operatorname{sen} \alpha \right)^2 \right)}{\cos^2 \alpha - \operatorname{sen}^2 \alpha} =$$

$$= \frac{2 \left(\frac{2}{4} \cos^2 \alpha - \frac{2}{4} \operatorname{sen}^2 \alpha \right)}{\cos^2 \alpha - \operatorname{sen}^2 \alpha} = \frac{\cos^2 \alpha - \operatorname{sen}^2 \alpha}{\cos^2 \alpha - \operatorname{sen}^2 \alpha} = 1.$$

$$c) \operatorname{sen} \alpha \cdot \cos 2\alpha - \cos \alpha \cdot \operatorname{sen} 2\alpha =$$

$$= \operatorname{sen} \alpha (\cos^2 \alpha - \operatorname{sen}^2 \alpha) - \cos \alpha \cdot 2 \operatorname{sen} \alpha \cos \alpha =$$

$$= \operatorname{sen} \alpha \cos^2 \alpha - \operatorname{sen}^3 \alpha - 2 \operatorname{sen} \alpha \cos^2 \alpha =$$

$$= \operatorname{sen} \alpha \cos^2 \alpha - \operatorname{sen}^3 \alpha = \operatorname{sen} \alpha (\cos^2 \alpha - \operatorname{sen}^2 \alpha) =$$

$$= \boxed{\operatorname{sen} \alpha \cdot \cos 2\alpha} \quad \downarrow \text{Factor común.}$$

$$d) \frac{2 \operatorname{tg} \frac{x}{2}}{1 + \operatorname{tg}^2 \frac{x}{2}} = \frac{2 \sqrt{\frac{1 - \cos x}{1 + \cos x}}}{1 + \left(\sqrt{\frac{1 - \cos x}{1 + \cos x}} \right)^2} = \frac{2 \sqrt{\frac{1 - \cos x}{1 + \cos x}}}{1 + \frac{1 - \cos x}{1 + \cos x}} =$$

$$= \frac{2 \sqrt{\frac{1 - \cos x}{1 + \cos x}}}{\frac{1 + \cos x + 1 - \cos x}{1 + \cos x}} = \frac{2 \sqrt{\frac{1 - \cos x}{1 + \cos x}}}{\frac{2}{1 + \cos x}} = \frac{2(1 + \cos x) \sqrt{\frac{1 - \cos x}{1 + \cos x}}}{2} =$$

$$= \sqrt{(1+\cos x)^2 \cdot \frac{1-\cos x}{1+\cos x}} = \sqrt{(1+\cos x)(1-\cos x)} = \sqrt{1-\cos^2 x} =$$

$$= \sqrt{\sin^2 x} = \underline{\underline{\sin x}}$$

13.-

a) $\sin(\alpha+\beta) \cdot \sin(\alpha-\beta) = (\sin \alpha \cos \beta + \sin \beta \cos \alpha) \cdot$

$(\sin \alpha \cos \beta - \sin \beta \cos \alpha) =$
 $\downarrow$
 soma x diferencia

$$= (\sin \alpha \cos \beta)^2 - (\sin \beta \cos \alpha)^2 = \sin^2 \alpha \cos^2 \beta - \sin^2 \beta \cos^2 \alpha =$$

$$= \sin^2 \alpha (1 - \sin^2 \beta) - \sin^2 \beta (1 - \sin^2 \alpha) =$$

$$= \sin^2 \alpha - \sin^2 \alpha \sin^2 \beta - \sin^2 \beta + \sin^2 \alpha \sin^2 \beta =$$

$$= \underline{\underline{\sin^2 \alpha - \sin^2 \beta}} \quad \checkmark$$

b) $\cos(\alpha+\beta) \cos(\alpha-\beta) = (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \cdot (\cos \alpha \cos \beta + \sin \alpha \sin \beta) =$

$$= (\cos \alpha \cos \beta)^2 - (\sin \alpha \sin \beta)^2 =$$

$$= \cos^2 \alpha \cos^2 \beta - \sin^2 \alpha \sin^2 \beta =$$

$$= \cos^2 \alpha (1 - \sin^2 \beta) - (1 - \cos^2 \alpha) \sin^2 \beta =$$

$$= \cos^2 \alpha - \cos^2 \alpha \sin^2 \beta - \sin^2 \beta + \cos^2 \alpha \sin^2 \beta = \underline{\underline{\cos^2 \alpha - \sin^2 \beta}} \quad \checkmark$$

c) $\sqrt{2} \cos\left(\frac{\pi}{4} - \alpha\right) = \sqrt{2} \cdot \left(\cos \frac{\pi}{4} \cos \alpha + \sin \frac{\pi}{4} \sin \alpha\right) =$

$$= \sqrt{2} \cdot \left(\frac{\sqrt{2}}{2} \cos \alpha + \frac{\sqrt{2}}{2} \sin \alpha\right) = \frac{\sqrt{2} \cdot \sqrt{2}}{2} \cos \alpha + \frac{\sqrt{2} \cdot \sqrt{2}}{2} \sin \alpha =$$

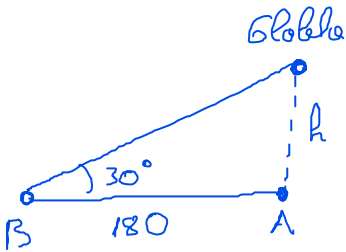
$$= \cos \alpha + \sin \alpha \quad \checkmark$$

$$\begin{aligned}
 d) \sin 3x &= \sin(x + 2x) = \sin x \cdot \cos 2x + \cos x \sin 2x = \\
 &= \sin x \cdot (\cos^2 x - \sin^2 x) + \cos x \cdot 2 \sin x \cos x = \\
 &= \sin x \cdot \cos^2 x - \sin^3 x + 2 \sin x \cos^2 x = \\
 &= 3 \sin x \cos^2 x - \sin^3 x \quad \checkmark
 \end{aligned}$$

q. e. d.
(quod erat demonstratum)

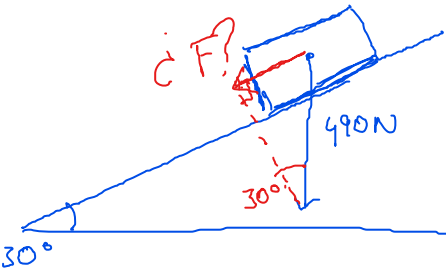
Soluciones - Repasados

14.-



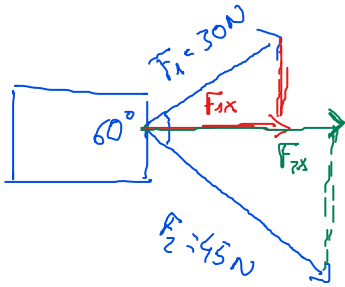
$$\tan 30^\circ = \frac{h}{180} \Rightarrow h = 180 \cdot \tan 30^\circ = 180 \cdot \frac{\sqrt{3}}{3} = \boxed{60\sqrt{3} \text{ m}}$$

15:



$$\sin 30^\circ = \frac{F}{490} \Rightarrow F = 490 \cdot \sin 30^\circ = 245 \text{ N}$$

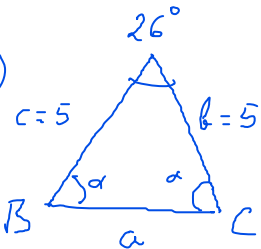
16. ~



Cada una de las fuerzas F_1 y F_2 que afectan al objeto generan una fuerza horizontal F_{1x} y F_{2x} . La fuerza total que afecta al objeto será la suma de F_{1x} y F_{2x} .

$$\begin{aligned} \bullet F_{1x} &= F_1 \cdot \cos 30^\circ = 30 \cdot \cos 30^\circ = 25'98\text{N} \\ \bullet F_{2x} &= F_2 \cdot \cos 45^\circ = 45 \cdot \cos 30^\circ = 38'97\text{N} \end{aligned} \left\{ \begin{array}{l} \rightarrow F = 25'98 + 38'97 = \\ = 64'95\text{N} \end{array} \right.$$

(7-)



Por el teorema del coseno $\rightarrow a^2 = b^2 + c^2 - 2bc \cdot \cos A$

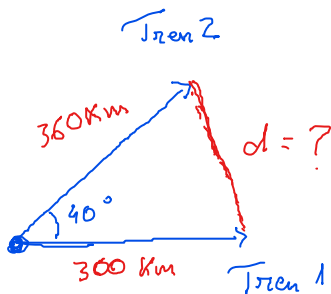
$$\rightarrow a^2 = 5^2 + 5^2 - 2 \cdot 5 \cdot 5 \cdot \cos 26^\circ = 5.06 \rightarrow a = \boxed{2.25 \text{ cm}}$$

Como B y C están en la misma $\rightarrow \alpha \Rightarrow \alpha + \alpha + 26^\circ = 180^\circ$

$$\Rightarrow 2\alpha + 26^\circ = 180^\circ \Rightarrow 2\alpha = 154^\circ \Rightarrow \alpha = 77^\circ$$

Por lo tanto $\boxed{B = C = 77^\circ}$

18



Tras dos horas de viaje cada tren ha recorrido:

- Трен 1 $\rightarrow 2 \cdot 150 = 300$ км

- Тлен 2 $\rightarrow 2 \cdot 180 = 360 \text{ Km}$

Para calcular d aplicamos el teorema del coseno $\Rightarrow$

$$d^2 = 300^2 + 360^2 - 2 \cdot 300 \cdot 360 \cdot \cos 40^\circ = 54134'$$

$$\Rightarrow d = \sqrt{51134'4} = \underline{\underline{223'67 \text{ km}}}$$

Los trenes están separados por $232'67 \text{ Km}$.

19. a) $a = 16 \text{ cm}$
 $b = 6 \text{ cm}$
 $A = 105^\circ$

$$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow \frac{16}{\sin 105^\circ} = \frac{6}{\sin B} \Rightarrow \sin B = \frac{6 \sin 105^\circ}{16} = \frac{3\sqrt{6} + 3\sqrt{2}}{32}$$

$B = 21'24''$
 $180^\circ - 21'24'' = 158'76''$ no valdría porque con A sumaría más de 180° .

$$C = 180^\circ - 105^\circ - 21'24'' = 53'76''$$

$$\frac{a}{\sin A} = \frac{c}{\sin C} \Rightarrow \frac{16}{\sin 105^\circ} = \frac{c}{\sin 53'76''} \Rightarrow c = \frac{16 \cdot \sin 53'76''}{\sin 105^\circ} = 13'36 \text{ cm}$$

b) $a = 5 \text{ cm}$
 $b = 6 \text{ cm}$
 $c = 11 \text{ cm}$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A \Rightarrow 5^2 = 6^2 + 11^2 - 2 \cdot 6 \cdot 11 \cdot \cos A \Rightarrow$$

$$\Rightarrow \cos A = \frac{-132}{-132} = 1 \Rightarrow A = 0^\circ \text{ } \underline{\text{imposible}} \rightarrow \text{Este triángulo no existe}$$

20. a) $1 + \frac{\sin^2 \alpha}{1 - \sin^2 \alpha} = 1 + \frac{\sin^2 \alpha}{\cos^2 \alpha} = 1 + \tan^2 \alpha = \frac{1}{\cos^2 \alpha} = \sec^2 \alpha$

$\sin^2 \alpha + \cos^2 \alpha = 1$
 $\cos^2 \alpha = 1 - \sin^2 \alpha$

b) $\frac{1 - \tan^2 x}{1 + \tan^2 x} = \frac{1 - \frac{\sin^2 x}{\cos^2 x}}{\frac{1}{\cos^2 x}} = \frac{\frac{\cos^2 x - \sin^2 x}{\cos^2 x}}{\frac{1}{\cos^2 x}} = \frac{(\cos^2 x - \sin^2 x) \cdot \cancel{\cos^2 x}}{\cancel{\cos^2 x} \cdot 1} = \cos^2 x - \sin^2 x = \cos 2x$

c) $\frac{\sin(a+b)}{\cos a \cdot \cos b} = \frac{\sin a \cdot \cos b + \cos a \cdot \sin b}{\cos a \cdot \cos b} = \frac{\sin a \cdot \cancel{\cos b}}{\cancel{\cos a} \cdot \cancel{\cos b}} + \frac{\cancel{\cos a} \cdot \sin b}{\cancel{\cos a} \cdot \cancel{\cos b}} =$

$$= \frac{\sin a}{\cos a} + \frac{\sin b}{\cos b} = \tan a + \tan b$$

d) $\cot(\alpha + \beta) = \frac{1}{\tan(\alpha + \beta)} = \frac{1}{\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta}} = \frac{1 - \tan \alpha \cdot \tan \beta}{\tan \alpha + \tan \beta} = \frac{1 - \frac{1}{\cot \alpha} \cdot \frac{1}{\cot \beta}}{\frac{1}{\cot \alpha} + \frac{1}{\cot \beta}} =$

$$= \frac{1 - \frac{1}{\cot \alpha \cdot \cot \beta}}{\frac{\cot \beta + \cot \alpha}{\cot \alpha \cdot \cot \beta}} = \frac{\frac{\cot \alpha \cdot \cot \beta - 1}{\cot \alpha \cdot \cot \beta}}{\frac{\cot \beta + \cot \alpha}{\cot \alpha \cdot \cot \beta}} = \frac{(\cot \alpha \cdot \cot \beta - 1)(\cancel{\cot \alpha \cdot \cot \beta})}{(\cancel{\cot \alpha \cdot \cot \beta}) \cdot (\cot \beta + \cot \alpha)} =$$

$$= \frac{\cot \alpha \cdot \cot \beta - 1}{\cot \alpha + \cot \beta}$$

q. e. d (quod erat demonstratum)